

PostgreSQL Backup and Restore Using pg_dumpall

PostgreSQL Backup and Restore Using pg_dumpall and psql

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Click to expand `pg_dumpall -help`

0. Introduction

Note: `pg_dumpall` does not support custom format backups.

The `pg_dumpall` utility is used to back up an entire PostgreSQL environment, including:

- * Roles and users
- * Tablespaces
- * All databases (schemas and data)

It is especially useful for:

- * Full cluster migrations
- * Disaster recovery
- * Environment replication across dev, QA, and prod

When restoring to a different host, you'll need to:

- * Update tablespace paths (using tools like `sed`)
- * Pre-create required tablespace directories

This ensures compatibility and successful restoration across different environments.

Backup

1. Backup ALL databases

```
[postgres@lxtrdpdsgv01 ~]$ nohup pg_dumpall -U postgres -v -f
/pgBackup/pgsql115/backup/backup_all_databases_10OCT2025.sql >
/pgBackup/pgsql115/backup/backup_all_databases_10OCT2025.log
2>&1 &
[1] 5610
[postgres@lxtrdpdsgv01 ~]$
```

```
[postgres@lxtrdpdsgv01 ~]$ cat
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.log |
grep -i
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql
pg_dumpall: running ""/usr/pgsql-15/bin/pg_dump" -v -f
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql -
Fa 'user=postgres dbname=template1'"
pg_dumpall: running ""/usr/pgsql-15/bin/pg_dump" -v -f
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql --
create -Fa 'user=postgres dbname=edpua'"
pg_dumpall: running ""/usr/pgsql-15/bin/pg_dump" -v -f
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql --
create -Fa 'user=postgres dbname=gebua'"
pg_dumpall: running ""/usr/pgsql-15/bin/pg_dump" -v -f
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql --
create -Fa 'user=postgres dbname=orcl'"
pg_dumpall: running ""/usr/pgsql-15/bin/pg_dump" -v -f
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql -
Fa 'user=postgres dbname=postgres'"
[postgres@lxtrdpdsgv01 ~]$
```

```
[postgres@lxtrdpdsgv01 ~]$ psql
psql (15.14)
Type "help" for help.
```

```
postgres=# \c orcl
You are now connected to database "orcl" as user "postgres".
```

```
orcl=# SELECT
orcl=#      schemaname || '.' || relname AS table_name,
orcl=#      pg_size_pretty(pg_total_relation_size(relid)) AS
total_size,
orcl=#      pg_size_pretty(pg_relation_size(relid)) AS
table_size,
orcl=#      pg_size_pretty(pg_total_relation_size(relid) -
pg_relation_size(relid)) AS index_size
orcl=# FROM
orcl=#      pg_catalog.pg_statio_user_tables
orcl=# ORDER BY
orcl=#      pg_total_relation_size(relid) DESC;
```

table_name	total_size	table_size	index_size
trd.metrics_high	799 MB	498 MB	301 MB
trd.metrics_mid	638 MB	398 MB	240 MB

```

trd.employees      | 493 MB      | 322 MB      | 172 MB
trd.sales_2023     | 399 MB      | 249 MB      | 150 MB
trd.sales_2024     | 399 MB      | 249 MB      | 150 MB
trd.sales_2022     | 399 MB      | 249 MB      | 150 MB
trd.sales_2021     | 398 MB      | 248 MB      | 150 MB
trd.metrics_low    | 159 MB      | 99 MB       | 60 MB
trd.test_data      | 71 MB       | 50 MB       | 21 MB
trd.emp_summary    | 24 kB       | 8192 bytes  | 16 kB
trd.metrics_rest   | 8192 bytes  | 0 bytes     | 8192 bytes
(11 rows)

```

```
orcl=#
```

2. Backup users/roles definition

```

[postgres@lxtrdpdsgv01 ~]$ nohup pg_dumpall -U postgres --
roles-only -v -f /pgBackup/pgsql15/backup/roles.sql >
/pgBackup/pgsql15/backup/roles.log 2>&1 &
[1] 5205
[postgres@lxtrdpdsgv01 ~]$

```

3. Backup tablespaces definition

```

[postgres@lxtrdpdsgv01 ~]$ nohup pg_dumpall -U postgres --
tablespaces-only -v -f
/pgBackup/pgsql15/backup/tablespaces.sql >
/pgBackup/pgsql15/backup/tablespaces.log 2>&1 &
[1] 5183
[postgres@lxtrdpdsgv01 ~]$

```

4. Backup dump only the schema, no data

```

[postgres@lxtrdpdsgv01 ~]$ nohup pg_dumpall -U postgres --
schema-only -v -f /pgBackup/pgsql15/backup/schemas.sql >
/pgBackup/pgsql15/backup/schemas.log 2>&1 &

```

```
[1] 4890  
[postgres@lxtrdpdsgv01 ~]$
```

5. Backup dump only the data, not the schema

```
[postgres@lxtrdpdsgv01 ~]$ nohup pg_dumpall -U postgres --  
data-only -v -f /pgBackup/pgsql15/backup/dataonly.sql >  
/pgBackup/pgsql15/backup/dataonly.log 2>&1 &  
[1] 5233  
[postgres@lxtrdpdsgv01 ~]$
```

6. Backup dump data as INSERT commands, rather than COPY

```
[postgres@lxtrdpdsgv01 ~]$ nohup pg_dumpall -U postgres --  
inserts -v -f /pgBackup/pgsql15/backup/inserts.sql >  
/pgBackup/pgsql15/backup/inserts.log 2>&1 &  
[1] 5274  
[postgres@lxtrdpdsgv01 ~]$
```

7. Backup dump only global objects, no databases

```
[postgres@lxtrdpdsgv01 ~]$ nohup pg_dumpall -U postgres --  
globals-only -v -f /pgBackup/pgsql15/backup/globals.sql >  
/pgBackup/pgsql15/backup/globals.log 2>&1 &  
[1] 5331  
[postgres@lxtrdpdsgv01 ~]$
```

Restore

8. Restore on same host

A. Drop Existing Databases (Optional)

Name	Owner	Encoding	Collate	Ctype
ICU Locale	Locale Provider	Access privileges		

```

-----+-----+-----+-----+-----
+-----+-----+-----+-----+-----
 postgres | postgres | UTF8      | en_US.UTF-8 | en_US.UTF-8 |
| libc      |
 template0 | postgres | UTF8      | en_US.UTF-8 | en_US.UTF-8 |
| libc      | =c/postgres      +
|           |           |           |           |
|           | postgres=CtC/postgres
 template1 | postgres | UTF8      | en_US.UTF-8 | en_US.UTF-8 |
| libc      | postgres=CtC/postgres+
|           |           |           |           |
|           | =c/postgres
(3 rows)

```

postgres=#

postgres=# \db

List of tablespaces

Name	Owner	Location
edpua_ix	postgres	/pgIx/pgsql15/edpua
edpua_tb	postgres	/pgTb/pgsql15/edpua
gebua_ix	postgres	/pgIx/pgsql15/gebua
gebua_tb	postgres	/pgTb/pgsql15/gebua
orcl_ix	postgres	/pgIx/pgsql15/orcl
orcl_tb	postgres	/pgTb/pgsql15/orcl
pg_default	postgres	
pg_global	postgres	

(8 rows)

postgres=#

B. Restore Full Backup

```

[postgres@lxtrdpdsgv01 ~]$ nohup psql -U postgres -X -f
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql -d
postgres >
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025_restor
e.log
[1] 4132

```

```
[postgres@lxtrdpdsgv01 ~]$

-- OR --

nohup psql -U postgres -f
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql >
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025_restore.log 2>&1 &

-- Log: backup_all_databases_10OCT2025_restore.log
```

C. Verification

```
[postgres@lxtrdpdsgv01 ~]$ psql
psql (15.14)
Type "help" for help.

postgres=# \l

                                List of
databases
   Name   | Owner   | Encoding | Collate  | Ctype    |
ICU Locale | Locale Provider | Access privileges
-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
  edpua   | postgres | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
| libc   |          |          |              |              |
  gebua   | postgres | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
| libc   |          |          |              |              |
  orcl    | postgres | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
| libc   |          |          |              |              |
 postgres | postgres | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
| libc   |          |          |              |              |
 template0 | postgres | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
| libc   |          | =c/postgres |              |              |
|          |          |          |              |              |
|          | postgres=CtC/postgres
 template1 | postgres | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
| libc   |          | postgres=CtC/postgres+
|          |          |          |              |              |
|          | =c/postgres
```



```
(6 rows)
```

```
postgres=#
```

9. Restore to Another Host (Different Tablespace Paths)

A. Copy Backup File

```
[postgres@lxtrdpdsgv01 backup]$ scp
backup_all_databases_10OCT2025.sql
192.168.2.31:/pgBackup/pgsql15/backup/
postgres@192.168.2.31's password:
backup_all_databases_10OCT2025.sql          100% 3556MB
51.5MB/s   01:09
[postgres@lxtrdpdsgv01 backup]$

[postgres@pg17 backup]$ ls -ltr
backup_all_databases_10OCT2025.sql
-rw-r--r--. 1 postgres postgres 3729147646 Oct  6 06:17
backup_all_databases_10OCT2025.sql
[postgres@pg17 backup]$
```

B. Extract Tablespace/Database Info

```
[postgres@lxtrdpdsgv01 backup]$ grep -i "CREATE TABLESPACE"
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql
CREATE TABLESPACE orcl_tb OWNER postgres LOCATION
'/pgTb/pgsql15/orcl';
CREATE TABLESPACE orcl_ix OWNER postgres LOCATION
'/pgIx/pgsql15/orcl';
CREATE TABLESPACE gebua_tb OWNER postgres LOCATION
'/pgTb/pgsql15/gebua';
CREATE TABLESPACE gebua_ix OWNER postgres LOCATION
'/pgIx/pgsql15/gebua';
CREATE TABLESPACE edpua_tb OWNER postgres LOCATION
'/pgTb/pgsql15/edpua';
CREATE TABLESPACE edpua_ix OWNER postgres LOCATION
'/pgIx/pgsql15/edpua';
```

```
[postgres@lxtrdpdsgv01 backup]$
[postgres@lxtrdpdsgv01 backup]$ grep -i "CREATE DATABASE"
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql
CREATE DATABASE edpua WITH TEMPLATE = template0 ENCODING =
'UTF8' LOCALE_PROVIDER = libc LOCALE = 'en_US.UTF-8'
TABLESPACE = edpua_tb;
CREATE DATABASE gebua WITH TEMPLATE = template0 ENCODING =
'UTF8' LOCALE_PROVIDER = libc LOCALE = 'en_US.UTF-8'
TABLESPACE = gebua_tb;
CREATE DATABASE orcl WITH TEMPLATE = template0 ENCODING =
'UTF8' LOCALE_PROVIDER = libc LOCALE = 'en_US.UTF-8'
TABLESPACE = orcl_tb;
[postgres@lxtrdpdsgv01 backup]$
```

C. Create Tablespace Directories on Target

```
[postgres@pg17 ~]$ psql
psql (15.13)
Type "help" for help.

postgres=# \l
                                     List of
databases
   Name   | Owner   | Encoding | Collate   | Ctype     |
ICU Locale | Locale Provider | Access privileges
-----+-----+-----+-----+-----+
 postgres | postgres | UTF8      | en_US.UTF-8 | en_US.UTF-8 |
| libc      |          |
 template0 | postgres | UTF8      | en_US.UTF-8 | en_US.UTF-8 |
| libc      | =c/postgres +
          |          |          |          |
          |          | postgres=CtC/postgres
 template1 | postgres | UTF8      | en_US.UTF-8 | en_US.UTF-8 |
| libc      | postgres=CtC/postgres+
          |          |          |          |
          |          | =c/postgres
(3 rows)

postgres=#
```

```

postgres=# \db
          List of tablespaces
   Name      |  Owner   | Location
-----+-----+-----
 pg_default  | postgres |
 pg_global   | postgres |
(2 rows)

postgres=#

```

Create Directory

```

[postgres@pg17 ~]$ mkdir -p /pgData/pgsql15/orcl_tb
[postgres@pg17 ~]$ mkdir -p /pgData/pgsql15/orcl_ix
[postgres@pg17 ~]$ mkdir -p /pgData/pgsql15/gebua_tb
[postgres@pg17 ~]$ mkdir -p /pgData/pgsql15/gebua_ix
[postgres@pg17 ~]$ mkdir -p /pgData/pgsql15/edpua_tb
[postgres@pg17 ~]$ mkdir -p /pgData/pgsql15/edpua_ix

```

D. Create Tablespaces in PostgreSQL

```

postgres=# CREATE TABLESPACE orcl_tb OWNER postgres LOCATION
'/pgData/pgsql15/orcl_tb';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE orcl_ix OWNER postgres LOCATION
'/pgData/pgsql15/orcl_ix';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE gebua_tb OWNER postgres LOCATION
'/pgData/pgsql15/gebua_tb';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE gebua_ix OWNER postgres LOCATION
'/pgData/pgsql15/gebua_ix';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE edpua_tb OWNER postgres LOCATION
'/pgData/pgsql15/edpua_tb';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE edpua_ix OWNER postgres LOCATION
'/pgData/pgsql15/edpua_ix';
CREATE TABLESPACE

```

```
postgres=#
```

```
-- OR ---
```

You can modify the tablespace paths directly in the backup file; however, this approach is not recommended for large backup files due to reliability concerns.

```
cd /pgBackup/pgsql15/backup/
```

```
# Replace ORCL paths
```

```
sed -i 's|/pgTb/pgsql15/orcl|/pgData/pgsql15/orcl_tb|g'
backup_all_databases_10OCT2025.sql
```

```
sed -i 's|/pgIx/pgsql15/orcl|/pgData/pgsql15/orcl_ix|g'
backup_all_databases_10OCT2025.sql
```

```
# Replace GEBUA paths
```

```
sed -i 's|/pgTb/pgsql15/gebua|/pgData/pgsql15/gebua_tb|g'
backup_all_databases_10OCT2025.sql
```

```
sed -i 's|/pgIx/pgsql15/gebua|/pgData/pgsql15/gebua_ix|g'
backup_all_databases_10OCT2025.sql
```

```
# Replace EDPUA paths
```

```
sed -i 's|/pgTb/pgsql15/edpua|/pgData/pgsql15/edpua_tb|g'
backup_all_databases_10OCT2025.sql
```

```
sed -i 's|/pgIx/pgsql15/edpua|/pgData/pgsql15/edpua_ix|g'
backup_all_databases_10OCT2025.sql
```

```
[postgres@pg17 ~]$ psql
```

```
psql (15.13)
```

```
Type "help" for help.
```

```
postgres=# \db
```

List of tablespaces

Name		Owner		Location
-----+-----+-----				
edpua_ix		postgres		/pgData/pgsql15/edpua_ix
edpua_tb		postgres		/pgData/pgsql15/edpua_tb
gebua_ix		postgres		/pgData/pgsql15/gebua_ix
gebua_tb		postgres		/pgData/pgsql15/gebua_tb
orcl_ix		postgres		/pgData/pgsql15/orcl_ix
orcl_tb		postgres		/pgData/pgsql15/orcl_tb
pg_default		postgres		

```
pg_global | postgres |  
(8 rows)  
  
postgres=#
```

E. Optional: Backup on Target Host

```
[postgres@pg17 ~]$ nohup pg_dumpall -U postgres -v -f  
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025_pg17.s  
ql >  
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025_pg17.l  
og 2>&1 &  
[1] 4246  
[postgres@pg17 ~]$
```

F. Restore on Target Host

```
[postgres@pg17 ~]$ nohup psql -U postgres -X -f  
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql -d  
postgres >  
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025_restor  
e_pg17.log 2>&1 &  
[1] 5213  
[postgres@pg17 ~]$
```

-- OR --

```
nohup psql -U postgres -f  
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025.sql >  
/pgBackup/pgsql15/backup/backup_all_databases_10OCT2025_restor  
e_pg17.log 2>&1 &
```

-- Log: backup_all_databases_10OCT2025_restore_pg17.log

G. Final Verification

```
[postgres@pg17 ~]$ psql
psql (15.13)
Type "help" for help.
```

```
postgres=# \l
```

```
List of databases
```

Name	Owner	Encoding	Collate	Ctype	ICU Locale	Locale Provider	Access privileges
edpua	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
gebug	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
orcl	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
postgres	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
template0	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
template1	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	

(6 rows)

```
postgres=#
```

```
[postgres@pg17 ~]$ psql
psql (15.13)
Type "help" for help.
```

```
postgres=# \c orcl
```

```
You are now connected to database "orcl" as user "postgres".
```

```
orcl=# SELECT
```

```
orcl=# schemaname || '.' || relname AS table_name,
```

```
orcl=# pg_size_pretty(pg_total_relation_size(relid)) AS
total_size,
```

```
orcl=# pg_size_pretty(pg_relation_size(relid)) AS table_size,
```

```
orcl=# pg_size_pretty(pg_total_relation_size(relid) -
pg_relation_size(relid)) AS index_size
```

```
orcl=# FROM
```

```
orcl=# pg_catalog.pg_statio_user_tables
```

```
orcl=# ORDER BY
```

```
orcl=# pg_total_relation_size(relid) DESC;
 table_name | total_size | table_size | index_size
-----+-----+-----+-----
trd.metrics_high | 799 MB | 498 MB | 301 MB
trd.metrics_mid | 638 MB | 398 MB | 240 MB
trd.employees | 493 MB | 322 MB | 172 MB
trd.sales_2023 | 399 MB | 249 MB | 150 MB
trd.sales_2024 | 399 MB | 249 MB | 150 MB
trd.sales_2022 | 399 MB | 249 MB | 150 MB
trd.sales_2021 | 398 MB | 248 MB | 150 MB
trd.metrics_low | 159 MB | 99 MB | 60 MB
trd.test_data | 71 MB | 50 MB | 21 MB
trd.emp_summary | 24 kB | 8192 bytes | 16 kB
trd.metrics_rest | 8192 bytes | 0 bytes | 8192 bytes
(11 rows)

orcl=#
```

Caution: *Your use of any information or materials on this website is entirely at your own risk. It is provided for educational purposes only. It has been tested internally, however, we do not guarantee that it will work for you. Ensure that you run it in your test environment before using.*

Thank you,

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